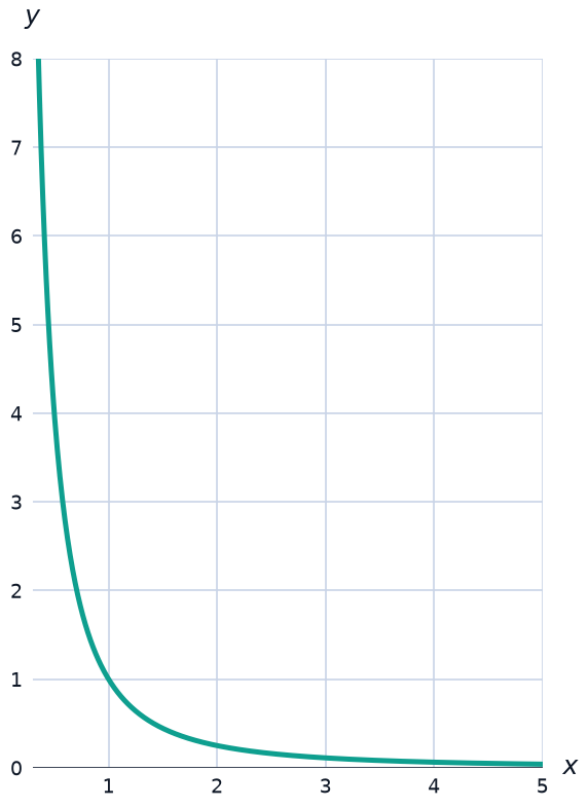
Partial Fractions and Improper Integrals

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

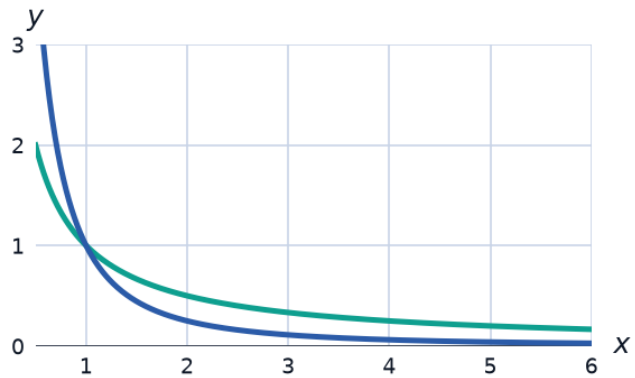
1 [Foundational] • **Target: Improper integrals: infinite limits**

Evaluate $\int_1^\infty \frac{1}{x^2}\,dx$, or show it diverges.



2 [Foundational] • **Target: Visualizing convergence: area under the curve**

By contrast, $\int_1^\infty \frac{1}{x}\,dx$ diverges. Compare the graphs of $y = \frac{1}{x}$ and $y = \frac{1}{x^2}$ on $[1, 10]$ and explain why $\frac{1}{x}$ decays too slowly for the area to stay finite.



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3 [Foundational] • Target: Partial fraction decomposition: distinct linear factors

Decompose $\frac{5x-1}{(x-1)(x+2)}$ into partial fractions.

4 [Foundational] • Target: Partial fraction decomposition: distinct linear factors

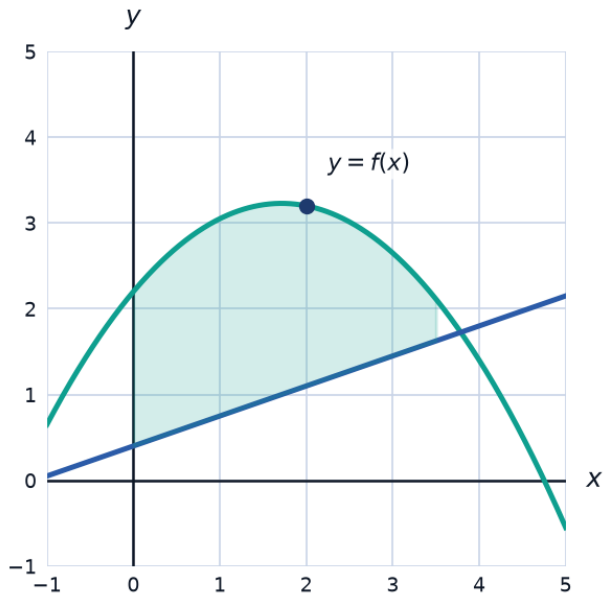
Evaluate $\int \frac{3x+5}{x^2+x-2}\,dx$ using partial fractions (note $x^2+x-2=(x+2)(x-1)$).

5 [Foundational] • Target: Partial fraction decomposition: distinct linear factors

Decompose $\frac{x+7}{x^2-x-6}$ into partial fractions (note $x^2-x-6=(x-3)(x+2)$).

6 [Foundational] • Target: Partial fractions with a repeated linear factor

Decompose $\frac{2x+3}{(x-1)^2}$ into partial fractions.



7 [Foundational] • Target: Partial fractions with a repeated linear factor

Evaluate $\int \frac{2x+3}{(x-1)^2}\,dx$ using the decomposition found above.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Partial fractions with an irreducible quadratic factor

Set up (but do not solve for constants) the partial fraction decomposition of $\frac{3x^2+2}{(x-1)(x^2+4)}$.

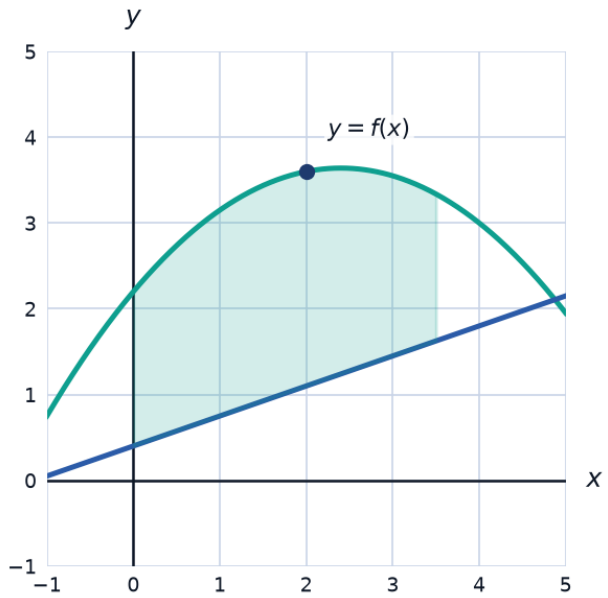
9 [Intermediate] • Target: Improper integrals: infinite limits

Evaluate $\int_1^{\infty} \frac{1}{x}\,dx$, or show it diverges.

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10 [Intermediate] • Target: Improper integrals: infinite limits

Evaluate $\int_0^\infty e^{-2x}\,dx$, or show it diverges.



11 [Intermediate] • Target: Improper integrals: infinite limits

Evaluate $\int_{-\infty}^0 e^x\,dx$, or show it diverges.

12 [Intermediate] • Target: Improper integrals: discontinuous integrands

Evaluate $\int_0^1 \frac{1}{\sqrt{x}}\,dx$, or show it diverges (note the integrand is undefined at $x = 0$).

13 [Intermediate] • Target: Improper integrals: discontinuous integrands

Evaluate $\int_0^1 \frac{1}{x}\,dx$, or show it diverges.

14 [Intermediate] • Target: Improper integrals: discontinuous integrands

Evaluate $\int_0^2 \frac{1}{(x-2)^2}\,dx$, or show it diverges (the integrand is undefined at $x = 2$).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

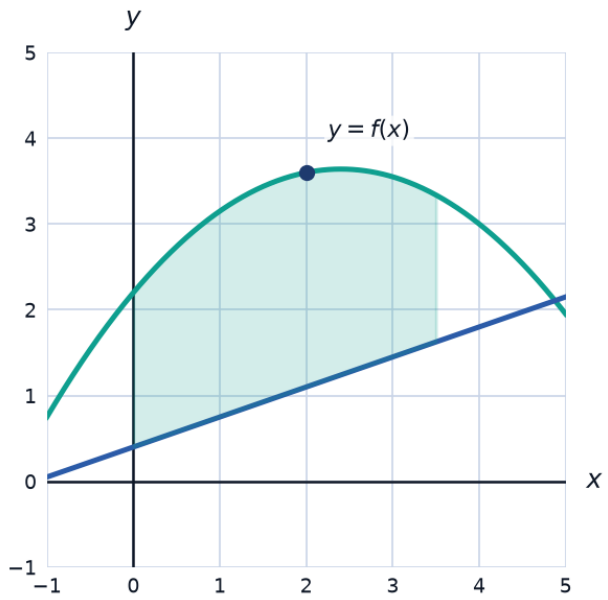
15 [Challenging] • Target: The p-test for improper integrals • Cross-Lesson: 4.02 Area in Polar Coordinates

State the p-test for $\int_1^\infty \frac{1}{x^p}\,dx$, and use it to determine convergence for $p = 3$ and $p = 0.5$.

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16 **[Challenging]** • **Target: The p-test for improper integrals** • **Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests**

State the analogous p-test for $\int_0^1 \frac{1}{x^p}\,dx$ (a discontinuity at $x = 0$), and determine convergence for $p = \frac{1}{2}$ and $p = 2$.



17 **[Challenging]** • **Target: Combining partial fractions with improper integrals** • **Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests**

Evaluate $\int_0^\infty \frac{1}{(x+1)(x+2)}\,dx$ using partial fractions.

18 **[Challenging]** • **Target: Three distinct linear factors** • **Cross-Lesson: 6.01 Power Series and Interval of Convergence**

Decompose $\frac{6x^2 - 3x - 1}{x(x-1)(x+1)}$ into partial fractions.

19 **[Challenging]** • **Target: Three distinct linear factors** • **Cross-Lesson: 6.02 Taylor and Maclaurin Series**

Evaluate $\int \frac{6x^2 - 3x - 1}{x(x-1)(x+1)}\,dx$ using the decomposition above.

20 **[Challenging]** • **Target: Visualizing convergence: area under the curve** • **Cross-Lesson: 1.01 Integration by Parts**

The convergent improper integral $\int_1^\infty \frac{1}{x^2}\,dx = 1$ (shown earlier) means the infinite region under $y = \frac{1}{x^2}$ for $x \geq 1$ has FINITE area. Explain intuitively why this is possible even though the region extends forever.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Partial fraction decomposition: distinct linear factors

For $\frac{7x+5}{(x+3)(x-4)} = \frac{A}{x+3} + \frac{B}{x-4}$, find A.

a. $-\frac{16}{7}$

b. $\frac{23}{7}$

c. $\frac{16}{7}$

d. $\frac{33}{7}$

22 [Foundational] • Target: Partial fractions with a repeated linear factor

For $\frac{7x-5}{(x)(x+5)} = \frac{A}{x} + \frac{B}{x+5}$, find A.

a. 8

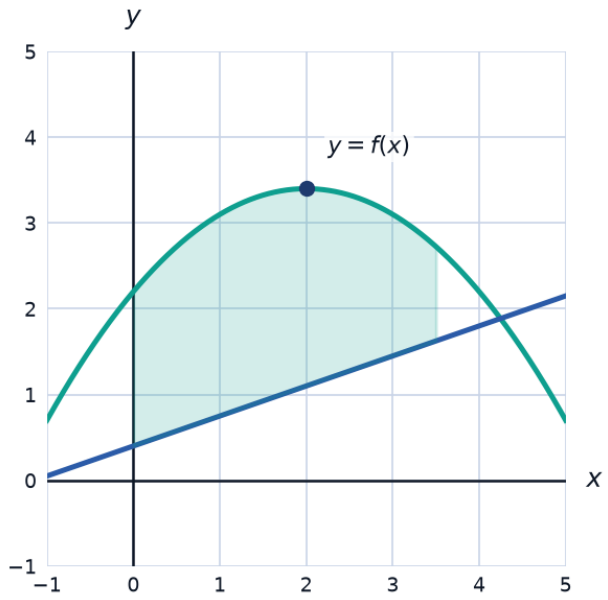
b. -1

c. 1

d. 0

23 [Foundational] • Target: Partial fractions with an irreducible quadratic factor

For $\frac{2x-6}{(x)(x+4)} = \frac{A}{x} + \frac{B}{x+4}$, find A.



a. $\frac{7}{2}$

b. $-\frac{3}{2}$

c. $\frac{3}{2}$

d. $-\frac{1}{2}$

24 [Foundational] • Target: Improper integrals: infinite limits

For $\frac{7x+1}{(x-3)(x-2)} = \frac{A}{x-3} + \frac{B}{x-2}$, find A.

a. 23

b. -22

c. -15

d. 22

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25 [Foundational] • Target: Improper integrals: discontinuous integrands

For $\frac{6x+9}{(x+1)(x-5)} = \frac{A}{x+1} + \frac{B}{x-5}$, find A.

a. $\frac{1}{2}$

b. $\frac{13}{2}$

c. $-\frac{1}{2}$

d. -1

26 [Foundational] • Target: The p-test for improper integrals

For $\frac{3x+3}{(x+3)(x-5)} = \frac{A}{x+3} + \frac{B}{x-5}$, find A.

a. $\frac{7}{4}$

b. $-\frac{3}{4}$

c. $\frac{9}{4}$

d. $\frac{3}{4}$

27 [Foundational] • Target: Combining partial fractions with improper integrals

For $\frac{3x-4}{(x-4)(x-5)} = \frac{A}{x-4} + \frac{B}{x-5}$, find A.

a. -8

b. 8

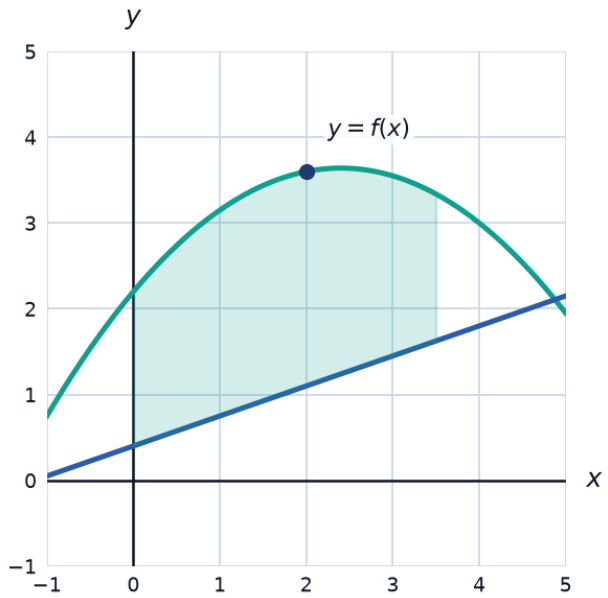
c. -7

d. 11

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28 [Foundational] • Target: Three distinct linear factors

Evaluate $\int_1^\infty \frac{2}{x^3}\,dx$, or show it diverges.



a. -1

b. 1

c. 2

d. $\frac{2}{3}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Visualizing convergence: area under the curve

Evaluate $\int_1^\infty \frac{8}{x^2}\,dx$, or show it diverges.

a. -8

b. 4

c. 8

d. 9

30 [Intermediate] • Target: Improper integrals with both an infinite bound and a discontinuity

Evaluate $\int_1^\infty \frac{3}{x^2}\,dx$, or show it diverges.

a. $\frac{3}{2}$

b. -3

c. 4

d. 3

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31 [Intermediate] • Target: Comparison test for improper integrals

Evaluate $\int_1^\infty \frac{4}{x^2}\,dx$, or show it diverges.

a. 5

b. -4

c. 2

d. 4

32 [Intermediate] • Target: Linear Partial Fractions

Evaluate $\int_1^\infty \frac{2}{x^4}\,dx$, or show it diverges.

a. $\frac{2}{3}$

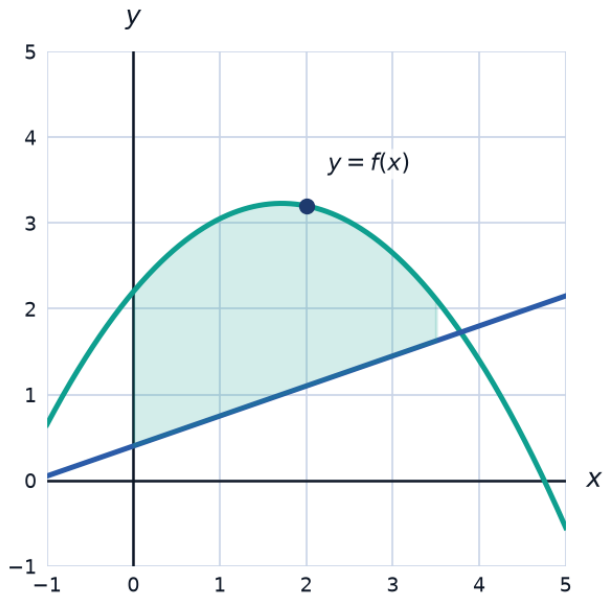
b. $\frac{5}{3}$

c. $\frac{1}{2}$

d. $-\frac{2}{3}$

33 [Intermediate] • Target: Repeated Linear & Irreducible Factors

Evaluate $\int_1^\infty \frac{8}{x^4}\,dx$, or show it diverges.



a. $\frac{8}{3}$

b. $-\frac{8}{3}$

c. $\frac{11}{3}$

d. 2

34 [Intermediate] • Target: Improper Integrals with Infinite Limits (Type 1)

Evaluate $\int_1^\infty \frac{3}{x^4}\,dx$, or show it diverges.

a. $\frac{3}{4}$

b. 1

c. 2

d. -1

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35 [Intermediate] • Target: Improper Integrals with Discontinuous Integrands (Type 2)

Evaluate $\int_0^1 \frac{2}{x^{\frac{1}{4}}}\,dx$, or show it diverges (the integrand is undefined at $x = 0$).

a. $\frac{8}{3}$

b. $\frac{11}{3}$

c. 8

d. $-\frac{8}{3}$

36 [Intermediate] • Target: Partial fraction decomposition: distinct linear factors

Evaluate $\int_0^1 \frac{6}{x^{\frac{1}{5}}}\,dx$, or show it diverges (the integrand is undefined at $x = 0$).

a. $\frac{15}{2}$

b. 30

c. $-\frac{15}{2}$

d. $\frac{17}{2}$

37 [Intermediate] • Target: Partial fractions with a repeated linear factor

For $\frac{5x-9}{(x-4)^2} = \frac{A}{x-4} + \frac{B}{(x-4)^2}$, find B.

a. 11

b. -4

c. 13

d. -11

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Partial fractions with an irreducible quadratic factor • Cross-Lesson: 6.01 Power Series and Interval of Convergence

For $\frac{5x-3}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$, find B.

a. -2

b. 4

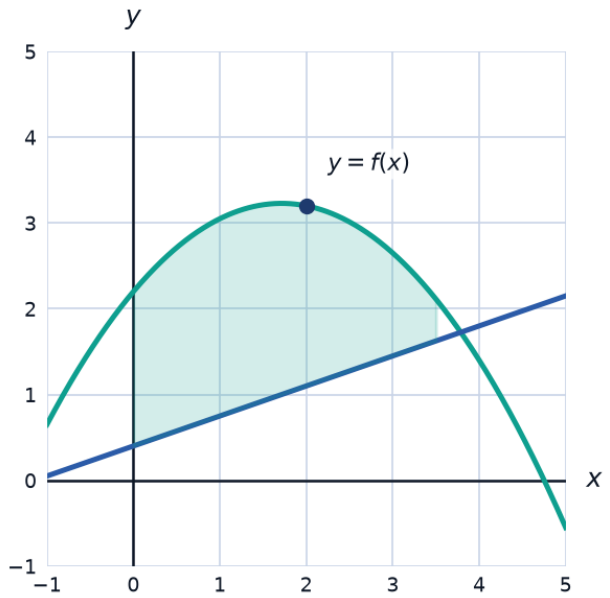
c. 6

d. 2

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39 [Challenging] • Target: Improper integrals: infinite limits • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Use a direct comparison to determine whether $\int_1^{\infty} \frac{2+\sin x}{x^2}\,dx$ converges or diverges.



a. Cannot be determined by comparison, since $\sin x$ oscillates.

b. Diverges, since $\frac{2+\sin x}{x^2} \geq \frac{1}{x^2}$ for $x \geq 1$, and $\int_1^{\infty} \frac{1}{x^2}\,dx$ diverges since $p = 2 \leq 1$.

c. Converges, since $\frac{2+\sin x}{x^2} \leq \frac{3}{x^2}$ for $x \geq 1$, and $\int_1^{\infty} \frac{3}{x^2}\,dx$ converges since $p = 2 > 1$.

d. Converges only because $a = 2$ happens to be even.

40 [Challenging] • Target: Improper integrals: discontinuous integrands • Cross-Lesson: 1.01 Integration by Parts

Use a direct comparison to determine whether $\int_1^{\infty} \frac{5+\sin x}{x}\,dx$ converges or diverges.

a. Diverges only because $a = 5$ happens to be even.

b. Diverges, since $\frac{5+\sin x}{x} \geq \frac{4}{x}$ for $x \geq 1$, and $\int_1^{\infty} \frac{4}{x}\,dx$ diverges since $p = 1 \leq 1$.

c. Cannot be determined by comparison, since $\sin x$ oscillates.

d. Converges, since $\frac{5+\sin x}{x} \leq \frac{6}{x}$ for $x \geq 1$, and $\int_1^{\infty} \frac{6}{x}\,dx$ converges since $p = 1 > 1$.

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41 [Challenging] • Target: Linear Partial Fractions • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Evaluate $\int \frac{1}{x^2-4}dx$.

a. $\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right|+C$

b. $\frac{1}{2}\ln\left|\frac{x-2}{x+2}\right|+C$

c. $\frac{1}{4}\ln|x^2-4|+C$

d. $\frac{1}{2}\arctan\left(\frac{x}{2}\right)+C$

42 [Challenging] • Target: Linear Partial Fractions • Cross-Lesson: 3.01 Parametric Equations

When must long division be performed BEFORE applying partial fractions?

a. When the degree of the numerator is greater than or equal to the degree of the denominator

b. When the denominator has repeated roots

c. When the numerator is a constant

d. Always

43 [Challenging] • Target: Repeated Linear & Irreducible Factors • Cross-Lesson: 3.02 Vector-Valued Functions

Evaluate $\int_2^3 \frac{1}{x(x-1)}dx$.

a. $\ln\left(\frac{4}{3}\right)$

b. $\ln 2$

c. $\ln 3$

d. $\frac{1}{6}$

44 [Challenging] • Target: Improper Integrals with Infinite Limits (Type 1) • Cross-Lesson: 4.01 Polar Curves and Derivatives

For what values of p does the improper integral $\int_1^\infty \frac{1}{x^p}dx$ converge?

a. $p > 1$

b. $p \geq 1$

c. $p < 1$

d. All real p

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45 [Challenging] • Target: Improper Integrals with Discontinuous Integrands (Type 2) • Cross-Lesson: 4.02 Area in Polar Coordinates

Evaluate the improper integral $\int_0^1 \frac{1}{\sqrt{x}}\,dx$.

a. 2

b. 1

c. ∞ (diverges)

d. $\frac{1}{2}$

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